Student number:	Seat No:

Mid-Semester Test – Sm1 2026
PRACTICE WITH ANSWERS

FNCE30007
Writing Time: 60 minutes
Reading Time: None
Weighting: 25%

Permitted Materials

- Calculator Casio FX82 (any suffix)
- Formula sheet provided.

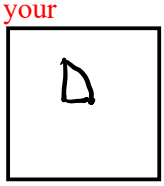
Instructions to Students

- During writing time, clearly write your student number and seat number on the cover sheet in the above boxes only. Ensure each digit is clearly written.
- Please write the answers in CAPITAL LETTERS only in the designated answer boxes provided within the exam paper for each question.
- Use a pen and not a pencil to write your answers
- Do not communicate with other students in the exam venue.
- No part of the exam paper may be removed from the venue.
- Ensure you only have standard permitted materials on your desk plus additional materials if listed above.
- Raise your hand if you require assistance.
- Students are NOT permitted to take pictures or make any reproduction via any means of any of the questions in the exam paper.

- Test contains 13 multiple-choice questions.
- Total marks: 15.
- Questions 4 and 5 are worth 2 marks each. All other questions are worth 1 mark.
- Select one answer to each question.

Question 1 (1 mark)
It is now 15th June 2020, you are bearish on the gold price for the next 6 months and willing to speculate by using $197,000 to take appropriate position in 6-month futures contract. The current spot price and December futures price are $1,950 and $1,975 per ounce respectively. Each contract is on 100 ounces of gold. The initial margin per contract required is $9,850. The maintenance margin per contract is $7,550. At the end of the first day of trading, the futures contract settles at $1,995 per ounce. Therefore:

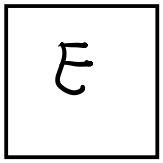
A. You must deposit $40,000 into your margin account to keep your position open.
B. You can withdraw $40,000 from your margin account.
C. You can withdraw $2,000 from your margin account.
D. You do not need to deposit into your margin account, but you cannot withdraw from your margin account either.
E. None of the other statements is correct



With $197,000, you are able to short 197,000/9850 = 20 contracts. The price has increased by $20. Because you have a short position you lose 20×100x20 or $40,000, i.e., $2000 per contract. The balance in the margin account therefore goes down from $197,000 to $157,000 which is above the maintenance margin of 7550*20 = $151,000. Hence you do not need to top up but you cannot withdraw either.

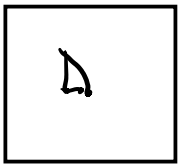
Question 2 (1 mark)
What does "basis risk" refer to in futures markets?

A) The risk of default by the futures exchange.
B) The difference between spot price and futures price.
C) The risk of regulatory changes impacting futures trading.
D) The risk of a company's financial health affecting futures contract prices.
E) The risk of the price relationship between the underlying asset and the futures contract changing over time in an unpredictable manner.



Question 3 (1 mark)
A farmer with wheat in the field takes a short position in wheat futures contracts expiring on November 15. The wheat underlying the futures is the same as the wheat the farmer will be selling. Which one of the following characterizes the farmer's position:

A. He faces basis risk if he's certain he can sell his wheat on November 15
B. If he makes a profit on his futures position, the farmer can eliminate basis risk
C. He is speculating on the price of wheat by taking a short futures position
D. He faces basis risk if he's not certain he can sell his wheat on November 15
E. None of the other statements is correct.



Question 4 (2 marks)
You just got hired as a consultant by a wealthy individual, Sir Topham Hat, who owns an equity portfolio currently worth $20,000,000. The beta of the portfolio is currently 1.1 but Sir Hat assesses that, given his forecast for the overall market, it is a good time to raise the portfolio risk exposure and wants to raise its beta to 1.8 for the period May 19th. – December 19th. The dividend yield on the S&P500 index is 1.5% per year and the current risk-free rate is 1% per year (both dividend yield and interest rates are expressed with continuous compounding). At the end of today, May 19th, the S&P500 spot closed at 750. You consider using E-Mini futures contract offered by CME on the S&P500, which have a multiplier of $50, and are available for March, June, September and December of each year.

Which of the following is the most appropriate strategy you would suggest Mr Hat to follow, given his goal for the portfolio beta?

a. Short 200 December contracts
b. Short 374 December contracts
c. Long 534 December contracts
d. Long 374 December contracts
e. Short 374 December contracts and Long 200 March contracts

D

S_t = 750 r=1% q=1.5% T=7/12 for December contract

F_t = S_t*exp[(r-q)*T] = 750*exp[(1%-1.5%)*7/12] = $747.816
$ Value of one Futures Contract is = $50*747.816 = $37,390.78

Fund should long (1.8 – 1.1)*20,000,000 / 37,390.78 = 374.42, i.e., 374 S&P500 futures contracts

Question 5 (2 marks)
On July 5, a stock index futures contract with expiration on December 20 settles at 357.24. On the same day, the spot index closes at 361.54. Assume that each index point is worth $1. The risk-free rate of interest is 1.5% per annum and the dividend yield on the spot index is 3.5% per annum (both rates are continuously compounded). Transaction costs are negligible. Then:

a. Go long the futures, short the spot, invest the proceeds and realize an arbitrage profit of $1.00
b. Go short the futures, borrow to long the spot and realize an arbitrage profit of $1.00
c. The futures and the spot are fairly priced and, hence, there is no arbitrage opportunity
d. Go long the futures, short the spot, invest the proceeds and realize an arbitrage profit of $4.30
e. One cannot tell if there is an arbitrage opportunity or not, as we don't know what the spot price of the index will be at the expiration of the futures contract

A

Answer:
S=361.54, r=1.5%, q=3.5%, T=5.5/12
No-arbitrage futures price = 361.54e^(1.5%-3.5%)*(5.5/12) = 358.241
Since the market price is lower than the no-arbitrage profit, hence it is too cheap. One should long futures, short spot, invest the proceeds, the arbitrage profits realized will be 358.241 - 357.24 = 1.00

Question 6 (1 mark)
Which of the following statements is NOT correct?

A. The optimal hedge ratio is chosen to minimize the variance of the hedged position
B. The optimal hedge ratio is always less than 1 if the hedge is not perfect
C. The optimal hedge ratio may not equal 1 even if changes in the spot price are perfectly positively correlated with changes in the futures price
D. When using index futures to hedge a short equity portfolio, a long futures position reduces the systematic risk of the hedged portfolio.
E. Asset mismatch and maturity mismatch affect the optimal hedge ratio

B

Question 7 (1 mark)
We watched the trading scene from the movie Trading Places. Which of the following statements is NOT correct:

a) Winthorpe and Valentine initially opened short futures positions and, after the futures price went down, they closed out by going long.
b) After the announcement by the Department of Agriculture, the vast majority of traders wanted to sell FCOJ (Frozen Concentrated Orange Juice) contracts to get out of the long positions they had accumulated.
c) Before the start of the trading session, Winthorpe and Valentine had inside information about the orange crop being large for the upcoming season.
d) The Dukes were asked by the clearinghouse to pay hundreds of millions in cash at the end of the trading session as they were net long and the price had decreased sharply during the session.
e) The Duke brothers thought the orange crop was going to be strong and, therefore, wanted to build a large short position in FCOJ futures.

E

Question 8 (1 mark)
During the lectures we covered the article "Cruise stocks slide as oil surges to $100 on Iran attacks". According to the article, which of the following statements is correct?

A. Cruise operators are naturally long fuel, so remaining unhedged is the safest choice.
B. Earnings of cruise operators that did not hedge are less vulnerable to increases in the cost of fuel.
C. Long futures hedge can reduce but not eliminate cost pressure if only part of a cruise operator's fuel needs is hedged.
D. A partial hedge by some of the cruise operators removes both price risk and quantity risk.
E. All major cruise operators adopted similar hedging strategies

C

Question 9 (1 mark)
Which of the following statements is true?

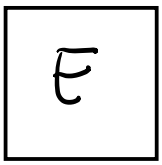
a) Forward contracts are traded on organized exchanges, but futures contracts are not.
b) Neither futures contracts nor forward contracts are traded on organized exchanges.
c) Both forward and futures contracts are traded on organized exchanges.
d) Forward contracts used to be traded on organized exchanges, but not anymore.
e) None of the above statements is true.

E

Question 10 (1 mark)

Which of the following statements is true?

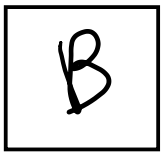
A. Gold and silver have always negative cost of carry
B. The 3-month no-arbitrage forward price of a consumption asset is solely determined by the spot price, interest rates over 3-months, and the income paid on the spot asset
C. Assume the convenience yield for copper is 1%. If the sum of financing cost and storage cost for copper over the next 6 months equals 5%, the 6-month no-arbitrage forward price of copper is 6% lower than the copper spot price
D. Assume the convenience yield for copper is 1%. If the sum of financing cost and storage cost for copper over the next 6 months equals 5%, the spot price of copper is 6% lower than the 6-month no-arbitrage forward price of copper
E. None of the other statements is true



Question 11 (1 mark)

An Australian importer knows it must pay EUR in 6 months. Management wants protection against the euro becoming more expensive but also wants to benefit if the euro ends up cheaper than expected. Which statement is most accurate?

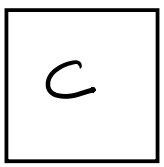
A. A forward contract and a euro call option both eliminate price risk and neither requires an upfront payment.
B. Buying a call option on euros can cap the worst-case purchase price while still allowing the firm to benefit if the euro becomes cheaper, but the option requires an upfront premium.
C. Selling euros forward is the best hedge because the firm is naturally short euros.
D. Writing a euro call option is the natural hedge because the firm will buy euros later.
E. A forward dominates an option because a forward also lets the firm benefit fully from favourable exchange-rate moves.



Question 12 (1 mark)

A non-dividend-paying stock is currently priced at $50. Two European call options on the stock have the same maturity. Option X has strike $50 and Option Y has strike $80. Which statement is most accurate?

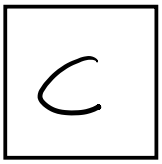
A. Option X and Option Y must have the same time value because they have the same maturity.
B. Option Y must have more time value because it is cheaper.
C. Option X is likely to have more time value because options near the money tend to have greater uncertainty about expiring in or out of the money.
D. Option Y must have a higher premium because it has more upside leverage.
E. European options have no time value because they can only be exercised at maturity.



Question 13 (1 mark)

A European call on a dividend-paying stock is priced at \$3.20. The stock price is \$41, the strike price is \$42, and the option expires in 6 months. Dividends of \$0.40 are expected in 2 months and 5 months. The continuously compounded risk-free rate is 8% p.a. What is the no-arbitrage price of a European put written on the same stock and with same strike and same maturity as the call?

A. \$2.11
B. \$2.78
C. \$3.33
D. \$4.05
E. \$4.74



Using put-call parity with dividends,

$$p = c + Ke^{-rT} + D - S$$

where $D$is the present value of dividends. Here,

$$p = 3.20 + 42e^{-0.08(0.5)} + 0.40e^{-0.08(2/12)} + 0.40e^{-0.08(5/12)} - 41 \approx 3.33$$

So, the put should be worth about \$3.33.